

Interpretable Knowledge Tracing with Difficulty-Aware Attention and Selective State Space Model

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Abstract

Knowledge Tracing (KT) aims to model students' knowledge states based on their historical learning sequence, playing a critical role in online education platforms. As the performance of sequence-based KT methods continues to improve, their increasing model complexity and lack of transparency have become significant limitations. In contrast, educational theory-driven KT methods incorporate educationally meaningful features (such as question difficulty or time spent on questions) to enhance interpretability and performance. However, these models typically adopt simpler structures to reduce complexity and avoid overfitting, which limits their ability to effectively capture the sequential characteristics of learning compared to sequence-based methods. To address these limitations, this paper aims to integrate the strengths of both types of methods by proposing an Interpretable KT approach with Difficulty-Aware Attention and Selective State Space Model (ASIKT). Specifically, leveraging educational context, we design a difficulty-enhanced attention mechanism to model students' knowledge retrieval process

at a fine-grained level without introducing additional parameters. Moreover, we employ selective state space model to capture the dynamic evolution of students' knowledge states based on their historical performance, ensuring both efficiency and simplicity in the model. Experimental results on four real-world datasets demonstrate that ASIKT outperforms state-of-the-art KT methods, achieving superior predictive performance. Additionally, ASIKT provides explanations for its predictions from multiple perspectives, showcasing strong interpretability. The code can be found at <https://github.com/mrsser/ASIKT>.

CCS Concepts

• **Applied computing** → **Education**; *Distance learning*; • **Social and professional topics** → **Student assessment**.

Keywords

knowledge tracing, question difficulty, user modeling

ACM Reference Format:

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1 Introduction

With the rapid advancement of internet technology, traditional educational methods are gradually being supplanted by emerging

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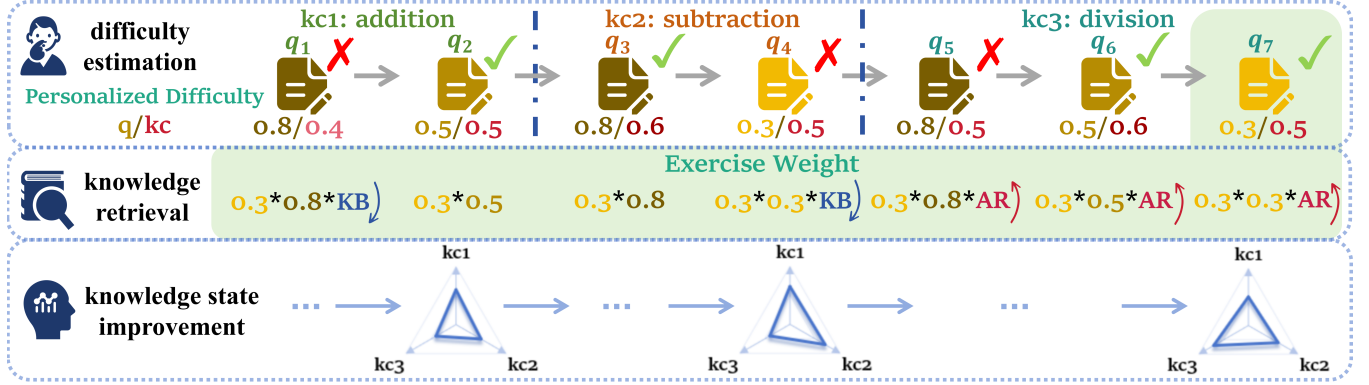


Figure 1: The student's question-solving process is divided into three stages. In the knowledge retrieval stage, *AR* refers to analogical reasoning, where students recall questions with the same KC to solve the current problem. *KB* stands for knowledge bias, meaning that if students have a sufficient knowledge state to answer the question correctly, they tend to rely on their past successful experiences. Conversely, if their knowledge state is weaker, they may be influenced by previous incorrect answers.

intelligent educational tools [2]. In recent years, the rise of online education platforms such as Massive Open Online Courses (MOOCs) [29] and Intelligent Tutoring Systems (ITS) [34] has led to the growing popularity of their convenient and digitalized educational services among both teachers and students [3]. Consequently, a key challenge has emerged: How to track students' learning progress based on their interactions and response records on online platforms, and accurately assess their knowledge state (i.e., the mastery of knowledge). This process, known as Knowledge Tracing (KT), has become a crucial technology for advancing intelligent education.

Figure 1 illustrates a simplified Knowledge Tracing process. In a student's question-answering sequence, each question is linked to a specific knowledge concept (KC). KT aims to model a student's knowledge state based on their response history and predict their likelihood of correctly answering the next question. Over the years, KT has evolved from traditional methods, such as the Hidden Markov Model [11], to more advanced sequence-based methods using Recurrent Neural Networks (RNNs) [24]) [33], and further to attention-based methods like SAKT [31], AKT [17], and others [10, 27, 43]. With increasing model complexity, sequence-based methods have achieved higher prediction accuracy. However, their lack of interpretability often limits their ability to provide insights aligned with practical educational needs, which poses challenges for broader adoption. In contrast, some studies incorporate educational meaningful features to improve both interpretability and performance. For instance, methods like LPKT [36] and HawkesKT [41] model forgetting mechanisms through response times and intervals, and methods like DIMKT [35] and QDCKT [25] explore the impact of question difficulty on students' question-solving processes. Although these approaches provide interpretable insights into modeling students' knowledge states, the inclusion of additional features often necessitates simpler model architectures to avoid overfitting and reduce computational complexity. As a result, these models may have a poor ability to capture the sequential characteristics of the learning compared to more advanced sequence-based methods. To address these limitations, our research aims to integrate the strengths of both types of knowledge tracing

methods. By leveraging advanced sequence modeling techniques to accurately trace students' knowledge states and incorporating educational meaningful factors, such as question difficulty, KC difficulty, and student behaviors like Slipping (i.e., having sufficient knowledge state but making a careless mistake) and Guessing (i.e., lacking sufficient knowledge state but guessing the correct answer), we aim to strike a balance between accuracy and interpretability, all while maintaining model simplicity.

As shown in Figure 1, based on real-world educational scenarios, a student's question-solving process can be divided into three stages: (1) Difficulty estimation stage: In this stage, we calculate personalized question difficulty using Bayesian estimation [7], accounting for students' subjective perceptions of difficulty. (2) Knowledge retrieval stage: In this stage, students recall relevant knowledge based on prior questions [5], where both question difficulty and the forgetting mechanism significantly shape the retrieval process. Specifically, more challenging questions tend to leave a deeper impression on students [6, 12]. Additionally, students exhibit phenomena such as analogical reasoning (AR) and knowledge bias (KB) [18, 22], emphasizing the importance of accurately modeling this knowledge retrieval process. Existing methods like DIMKT and QDCKT overlook this process, while AKT attempts to address it using monotonic attention. However, AKT relies on a simplistic IRT model [15] to incorporate question difficulty, which limits its effectiveness. Therefore, we introduce Exercise Weight, derived from question difficulty, AR, and KB, as well as Memory Decay, modeled using the forgetting curve [14], to enhance the attention mechanism. These enhancements enable the model to effectively capture the impact of question difficulty on the student's knowledge retrieval process while leveraging historical information in the learning sequence. (3) Knowledge state improvement stage: In this stage, students update their knowledge based on their performance. To model these updates, it is crucial to account for performance across multiple time steps to accurately capture the dynamic changes in their knowledge state. While methods incorporating educational features [25, 35, 36] consider only the influence of the immediate prior time step, sequence-based methods, such as those utilizing

Transformer [40] architectures [17, 27, 43] analyze the entire learning sequence. However, these methods often suffer from increased complexity and limited interpretability. To this end, we adopt the selective state space model (SSM) [19], which effectively models the influence of each time step. Our results demonstrate that SSM captures the dynamics of students' knowledge states while maintaining a simple and interpretable structure, making it a compelling choice for knowledge tracing.

In this paper, we propose a novel Interpretable **KT** method with Difficulty-Aware Attention and Selective SSM (ASIKT). ASIKT effectively traces students' knowledge states by incorporating question difficulty, enhancing interpretability without increasing model complexity. First, we compute personalized question difficulty for each student based on their learning history. We then design the Difficulty-Aware Knowledge Retriever (DAKR) module, which integrates Exercise Weight and Memory Decay into the dot-product attention mechanism, refining the knowledge retrieval process while maintaining computational efficiency. Next, we propose the Knowledge State Modeler (KSM) module, which uses selective SSM to capture dynamic changes in knowledge states. Compared to complex sequence-based methods, SSM efficiently learns from historical sequence information, achieving a balance between simplicity and accuracy. Finally, the prediction module considers Slipping and Guessing behaviors to forecast students' future responses. Experimental results on four real-world datasets show that ASIKT outperforms state-of-the-art KT methods, demonstrating superior performance. Moreover, ASIKT offers strong interpretability by providing insightful explanations for the learned knowledge states and predictions, taking into account multiple factors, including question and KC difficulty, as well as Slipping and Guessing behaviors. Our main contributions can be summarized as follows:

- We propose an attention mechanism enhanced with Exercise Weight and Memory Decay to finely model the student knowledge retrieval process, improving the model's learning ability and interpretability without introducing additional parameters.
- We employ a selective SSM to capture the dynamic changes in students' knowledge states based on their historical performance, enhancing accuracy while maintaining the model's efficiency and simplicity.
- Extensive experimental results validate the effectiveness of ASIKT. Furthermore, ASIKT provides reasonable explanations for the model's learned outcomes from multiple dimensions, including question and KC difficulty, Slipping, and Guessing, demonstrating strong interpretability.

2 Related Work

In this section, we first introduce sequence-based KT methods in chronological order, followed by a classification of KT methods based on educational features.

2.1 Sequence-based KT Methods

With the rapid development of deep learning, sequence-based knowledge tracing methods have gradually replaced traditional probabilistic reasoning-based methods [11] and factor analysis-based

methods [9, 32], becoming the mainstream approach in knowledge tracing research. DKT [33] was the first to introduce deep learning to KT, utilizing RNNs [24] and LSTMs [20] to learn from students' question-solving sequences. Building on sequence modeling, DKVMN [44] and SKVMN [1] integrated a key-value memory structure into the network. With the re-design of memory retrieval mechanisms in Transformer [40], methods like SAKT [31] and SAINT [10] applied attention mechanisms to KT, leading to significant advancements. Moreover, AKT [17] employs a monotonic attention mechanism to model the forgetting behavior in students' question-solving processes. Recent advancements in KT methods include simpleKT [27], which simplifies the model architecture based on AKT while maintaining performance; DTransformer [43], which combines contrastive learning and Transformer structures for knowledge state tracking; and Mamba4KT [8], which utilizes SSM [19] to improve training efficiency and resource utilization.

2.2 KT Methods with Educational Features

As the complexity of sequence-based models continues to increase, some studies leverage educationally meaningful features to enhance model interpretability. For instance, methods like GKT [30] and MLC-DKT [45] leverage graph structures and question-KC associations to explore the topological relationships between KCs and questions. Additionally, studies like HawkesKT [41] and LPKT [36] incorporate time-related factors, such as duration and intervals of interaction, to model forgetting behaviors and further refine the tracking of knowledge state. However, not all datasets provide the additional features mentioned above. In contrast, question difficulty, which can be directly derived from dataset statistics, has gradually gained attention in KT studies. PEBG [26] enhances question embeddings by incorporating question difficulty levels. DIMKT [35] explores the impact of question difficulty on students' knowledge states through a specialized network structure. TCKT [21] models the causal relationships between knowledge concepts while leveraging time-related factors and question difficulty to simulate knowledge acquisition. Furthermore, recent knowledge tracing research has increasingly focused on designing models based on educational context. QDCKT [25] replaces question embeddings with question difficulty to explore the consistency between question difficulty and knowledge states. PKT [37] applies a stage-based modeling approach to capture students' learning processes and investigates the impact of question difficulty based on IRT. MIKT [38] tracks students' knowledge states from both coarse-grained and fine-grained perspectives, while extending the Rasch model to study the influence of question difficulty.

In summary, while recent sequence-based methods have improved performance with deep learning, they come with increased complexity. Although Mamba4KT [8] was the first to apply SSM to simplify KT models, it only replaced the attention mechanism in AKT with SSM, without fully exploring its potential for KT. Methods using question difficulty, like DIMKT and QDCKT, adopt simpler structures to avoid overfitting, which limits their learning capacity. In contrast, our research enhances attention mechanisms with educational context and utilizes selective SSM to model students' knowledge retrieval processes, reducing complexity while boosting interpretability.

3 Preliminary

In this section, we define the knowledge tracing problem and the concept of question and KC difficulty, including both global and personalized difficulty.

3.1 Problem Definition

In an intelligent tutoring system, assume that there is a student set $S = \{s_1, s_2, \dots, s_I\}$ containing I distinct students, an question set $Q = \{q_1, q_2, \dots, q_J\}$ containing J distinct questions, and a knowledge concept set $KC = \{kc_1, kc_2, \dots, kc_M\}$ containing M distinct knowledge concepts. Given the historical sequence $X_t = \{(q_1, kc_1, a_1), (q_2, kc_2, a_2), \dots, (q_t, kc_t, a_t)\}$ of a particular student, where $a_t \in \{0, 1\}$ represents the student's answer (1 for correct, 0 for incorrect), knowledge tracing aims to track the evolution of the student's knowledge state throughout the question-solving process. It predicts the probability $p(a_{t+1}|X_t, q_{t+1}, kc_{t+1})$ that the student will answer the question q_{t+1} correctly at time $t + 1$, based on their current knowledge state at time t .

3.2 Question and KC Difficulty Definition

To capture the personalized perception of difficulty, we define both global and personalized difficulty for questions and KCs in a unified manner. Let e denote either a question or a KC. Global difficulty reflects the inherent difficulty of e from a population-level perspective. Due to data sparsity, direct accuracy estimation may be biased. Thus, following the Bayesian smoothing approach inspired by TCKT [21], we define the global difficulty of e as:

$$GD_e = 1 - \frac{A(D, e) \cdot C(D, e) + \bar{A} \cdot \frac{|D|}{|E|}}{C(D, e) + \frac{|D|}{|E|}}, \quad (1)$$

where D represents the collection of all question-solving records, $C(D, e)$ is the count of e in D , $A(D, e)$ is its empirical accuracy, $|E|$ is the total number of distinct elements (questions or KCs), and $\bar{A} = \frac{1}{|E|} \sum_{i=1}^{|E|} A(D, e_i)$ denotes the average accuracy across all e_i .

Personalized difficulty reflects how a student perceives the difficulty of e based on their past interactions. Let X_{t-1} denote a student's question-solving sequence up to time $t - 1$. Then, the personalized difficulty of e_t is computed as:

$$PD_{e_t} = \frac{(1 - A(X_{t-1}, e_t)) \cdot C(X_{t-1}, e_t) + GD_{e_t} \cdot C(D, e_t)}{C(X_{t-1}, e_t) + C(D, e_t)}, \quad (2)$$

where $A(X_{t-1}, e_t)$ and $C(X_{t-1}, e_t)$ denote the student's accuracy and count for e_t in their past sequence. This formulation combines personal and global difficulty estimates to balance student-specific signals and data sparsity.

4 The ASIKT Model

In this section, we provide a detailed introduction to the proposed ASIKT model, with its architecture illustrated in Figure 2. First, we embed the questions and KCs from the input sequence. Next, we present the Difficulty-Aware Knowledge Retriever (DAKR), which uses an attention mechanism enhanced by Exercise Weight and Memory Decay to capture the impact of question difficulty on students' question-solving processes. We then introduce the Knowledge State Modeler (KSM), which applies selective SSM to effectively

model the evolution of students' knowledge states. Finally, the Prediction Network forecasts the probability of a correct answer at the next time step based on the students' current knowledge state, while accounting for their Slipping & Guessing behaviors.

4.1 Embedding Layer

The input to the ASIKT model consists of the student's historical question-solving sequence X_t , along with the question q_{t+1} and its corresponding knowledge concept kc_{t+1} at the next time step. At time t , the question embedding $\mathbf{q}_t \in \mathbb{R}^d$, KC embedding $\mathbf{kc}_t \in \mathbb{R}^d$, and answer embedding $\mathbf{a}_t \in \mathbb{R}^d$ are obtained through the embedding matrices $\mathbf{Q} \in \mathbb{R}^{J \times d}$, $\mathbf{KC} \in \mathbb{R}^{M \times d}$, and $\mathbf{A} \in \mathbb{R}^{2 \times d}$, respectively. Here, J and M represent the number of distinct questions and KCs in the dataset, respectively, while d denotes the embedding dimension. To capture the impact of historical answers on the student's knowledge state, we further embed the question-answer pairs (q_t, a_t) and KC-answer pairs (kc_t, a_t) as follows: $\mathbf{qa}_t = \mathbf{q}_t + \mathbf{a}_t$, $\mathbf{qa}_t \in \mathbb{R}^d$ and $\mathbf{kca}_t = \mathbf{kc}_t + \mathbf{a}_t$, $\mathbf{kca}_t \in \mathbb{R}^d$. Finally, we obtain the embedding sequences that include the information of the next time step: $\mathbf{q}_{2:t+1}$ and $\mathbf{kc}_{2:t+1}$, as well as the embedding sequences representing the historical answer information: $\mathbf{qa}_{1:t}$ and $\mathbf{kca}_{1:t}$.

4.2 Difficulty-Aware Knowledge Retriever

In the DAKR block, we examine the fine-grained impact of question and KC difficulty on the input embedding sequences $\mathbf{q}_{2:t+1}$, $\mathbf{kc}_{2:t+1}$, $\mathbf{qa}_{1:t}$ and $\mathbf{kca}_{1:t}$, respectively. We first compute Exercise Weight based on personalized difficulty, which serves as an attention coefficient for students' retrieval processes. Next, we calculate Memory Decay using the Forgetting Curve [14] to model forgetting mechanisms. Finally, both Exercise Weight and Memory Decay are integrated into the dot-product attention, guiding the model to prioritize recent or more difficult questions (or KCs), simulating knowledge retrieval without adding model complexity.

Exercise Weight. For the input embedding sequences $\mathbf{q}_{2:t+1}$ and $\mathbf{kc}_{2:t+1}$, which do not include answer information, the Exercise Weight between q_t and q_τ can be computed as follows:

$$w_{t,\tau}^{(na)} = \begin{cases} PD_{q_t} \cdot PD_{q_\tau} \cdot AR_t & \text{if } kc_t = kc_\tau, \\ PD_{q_t} \cdot PD_{q_\tau} & \text{if } kc_t \neq kc_\tau. \end{cases} \quad (3)$$

For the input embedding sequences $\mathbf{qa}_{1:t}$ and $\mathbf{kca}_{1:t+1}$, which include answer information, the Exercise Weight can be computed as:

$$w_{t,\tau}^{(a)} = \begin{cases} PD_{q_t} \cdot PD_{q_\tau} \cdot AR_t & \text{if } kc_t = kc_\tau, \\ PD_{q_t} \cdot PD_{q_\tau} & \text{if } kc_t \neq kc_\tau \text{ and } a_t = a_\tau, \\ PD_{q_t} \cdot PD_{q_\tau} \cdot KB & \text{if } kc_t \neq kc_\tau \text{ and } a_t \neq a_\tau. \end{cases} \quad (4)$$

Here $\tau \leq t$, $w_{t,\tau}^{(na)}$ or $w_{t,\tau}^{(a)}$ represent the attention weight that a student recalls for the question (or KC) at historical time τ when answering the question at the current time t . The gain factor AR_t , define as $AR_t = \exp(PD_{kc_t})$, with $AR > 1$, amplifies the influence of difficulty for questions sharing the same KC. A higher personalized difficulty of the KC associated with the question result in a larger AR . KB is a decay factor for questions associated with different KCs and varying answers. We define $KB = 0.5$. These factors respectively

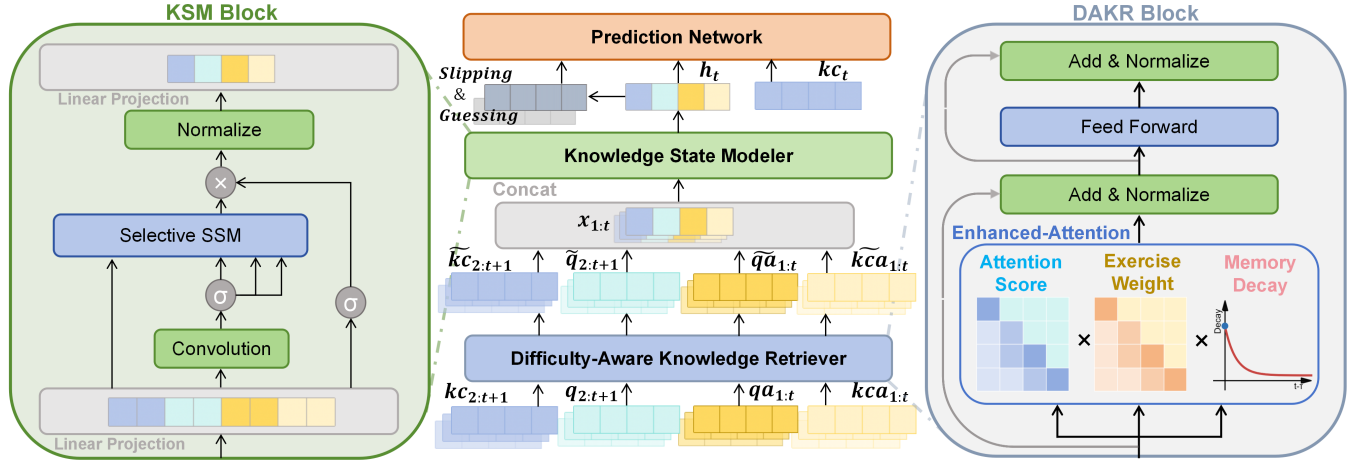


Figure 2: The architecture of the ASIKT model.

simulate students analogical reasoning behavior [18] through *AR* and highlight the phenomenon of knowledge bias [22] using *KB*.

Memory Decay. The student's forgetting mechanism also plays a significant role in their knowledge retrieval process. Related research has shown that exponential functions, with their relatively simple form, are best suited to modeling the decay of memory over time [4]. Therefore, we define the Memory Decay factor $m_{t,\tau}$ as follows:

$$m_{t,\tau} = \lambda + (1 - \lambda) * \eta * \exp(-\xi * (t - \tau)). \quad (5)$$

Here, $\lambda \in (0, 1)$ is a learnable parameter representing the minimum probability of memory retention. $\eta \leq 1$ is a constant that denotes the initial probability of memory retention immediately after learning; in our experiments, $\eta = 1$. $\xi > 0$ is another learnable parameter that determines the rate at which memory decays over time. $m_{t,\tau}$ quantifies the extent to which a student forgets the question (or KC) at historical time τ when answering a question at the current time t .

The Enhanced Attention Mechanism. To explore the impact of question difficulty and the forgetting mechanism on the student's knowledge retrieval process, we modify the original attention mechanism as follows:

$$s_{t,\tau} = \frac{q_t^{(attn)} \cdot k_\tau^T}{\sqrt{d_k}} * w_{t,\tau} * m_{t,\tau}. \quad (6)$$

The modified attention score $s_{t,\tau}$ is obtained by multiplying the original dot-product attention score with the Exercise Weight $w_{t,\tau}$ and Memory Decay factor $m_{t,\tau}$. When the input sequence is $q_{2:t+1}$ or $kc_{2:t+1}$, $w_{t,\tau} = w_{t,\tau}^{(na)}$. Otherwise, $w_{t,\tau} = w_{t,\tau}^{(a)}$. After applying the Softmax operation, we obtain the final enhanced attention score as:

$$\alpha_{t,\tau} = \frac{\exp(s_{t,\tau})}{\sum_{\tau'}^t \exp(s_{t,\tau'})}. \quad (7)$$

Under the above settings, the enhanced attention mechanism can further focus on questions (or KCs) with greater Exercise Weight, thereby improving the model's interpretability without increasing its complexity. At the same time, it can automatically learn the

student's forgetting mechanism through Memory Decay, enabling fine-grained modeling of the student's knowledge retrieval process.

The Overall DAKR Structure. For the input embedding sequences $q_{2:t+1}$, $kc_{2:t+1}$, $qa_{1:t}$ and $kca_{1:t}$, we learn each of them individually and ultimately output enhanced sequences $\tilde{q}_{2:t+1}$, $\tilde{kc}_{2:t+1}$, $\tilde{qa}_{1:t}$ and $\tilde{kca}_{1:t}$, which contain contextual information. Specifically, we apply the enhanced attention along with the multi-head mechanism. Notably, each head learns a different Memory Decay factor. Following the encoder design in Transformer [40], the outputs of each head are concatenated into a single vector and passed through subsequent sub-layers, including residual connections, layer normalizations, dropout layers, and feed-forward neural network.

4.3 Knowledge State Modeler

After fine-grained learning of the student's knowledge retrieval process, the KSM is proposed to model the student's knowledge state. We apply selective SSM to learn from the student's historical learning sequence, accurately capturing changes in their knowledge state while maintaining the simplicity of the model.

Selective State Space Model. We employ the selective SSM from the Mamba structure, as proposed by [13]. For the input discrete sequence $x_{1:t} \in \mathbb{R}^{t \times d_h}$, the model is expressed as follows:

$$h_t^{(SSM)} = A_t h_{t-1}^{(SSM)} + B_t x_t, \quad (8)$$

$$y_t = C_t^T h_t^{(SSM)}, \quad (9)$$

where A_t is a constant obtained from the input x_t ; both $B_t \in \mathbb{R}^{d_s}$ and $C_t \in \mathbb{R}^{d_s}$ are vectors derived from the mapping of the input x_t . By eliminating the hidden state $h_t^{(SSM)}$ in the above equations, we can directly maps the input x to the output y as follows:

$$y_t = A_{\tau:t}^x \sum_{\tau}^t C_\tau^T B_\tau x_\tau, \quad (10)$$

$$y = \text{SSM}(A, B, C)(x) = L \cdot x. \quad (11)$$

Here, $\mathbf{L} \in \mathbb{R}^{t \times t}$ is a lower triangular matrix, with elements $\mathbf{l}_{j,i} = A_{i,j}^\times \cdot (\mathbf{C}_i^T \mathbf{B}_j)$, where $A_{i,j}^\times = A_i A_{i+1} \cdots A_{j+1}$. These elements represent the influence weight of the historical time step τ on the current time step t in the input sequence.

The Overall KSM Structure. In the KSM block, we first concatenate the outputs from the DAKR block as follows: $\mathbf{x}_t = \tilde{\mathbf{q}}_{t+1} \oplus \tilde{\mathbf{k}}_{t+1} \oplus \tilde{\mathbf{q}}_t \oplus \mathbf{k} \mathbf{c}_t$, forming $\mathbf{x} = \mathbf{x}_{1:t}$, $\mathbf{x} \in \mathbb{R}^{t \times (4 \times d)}$ as the input sequence. This sequence is first passed through a linear layer to generate constant $\mathbf{A}$, followed by a convolutional layer, and an SiLU activation function before being fed into the selective SSM. The SSM is then employed to learn the hidden information within the sequence, as illustrated below:

$$\mathbf{A} = \text{Linear}(\mathbf{x}), \quad (12)$$

$$\mathbf{x}', \mathbf{B}, \mathbf{C} = \text{SiLU}(\text{Conv1D}(\text{Linear}(\mathbf{x}))), \quad (13)$$

$$\mathbf{y} = \text{SSM}(\mathbf{A}, \mathbf{B}, \mathbf{C})(\mathbf{x}') = \mathbf{L} \cdot \mathbf{x}'. \quad (14)$$

Similar to the multi-head self-attention mechanism, the SSM also employs a multi-head mechanism to learn the input sequence across different time scales, with the outputs ultimately concatenated to form $\mathbf{y}$. Finally, the output $\mathbf{y}$ from the SSM is processed through residual connections, layer normalization, and a linear layer to derive the student's knowledge state $\mathbf{h} = \mathbf{h}_{1:t}$, $\mathbf{h} \in \mathbb{R}^{t \times (4 \times d)}$:

$$\mathbf{h} = \text{Linear}(\text{LayerNorm}(\mathbf{y} \cdot \text{SiLU}(\text{Linear}(\mathbf{x}))))). \quad (15)$$

As shown in Equation 10, the student's knowledge state $\mathbf{h}_t$ at time step t can be interpreted as a weighted sum of the exercise information from each historical time step, ranging from 1 to $t - 1$. This indicates that the SSM modeling process for the student's knowledge state is highly interpretable.

4.4 Prediction and Objective Function

After modeling the student's knowledge state, we predict the probability that the student will correctly answer the question q_{t+1} at the next time step $t + 1$. In this process, we consider not only the student's current knowledge state $\mathbf{h}_t$, but also the influence of the student's Slipping and Guessing behaviors on the final response. Specifically, the probabilities of Slipping (p_s) and Guessing (p_g), as well as the probability based on the knowledge state (p_k), can be obtained separately by concatenating $\mathbf{h}_t$ with the current KC embedding $\mathbf{k} \mathbf{c}_t$, followed by fully connected layers and the sigmoid activation functions. Then, the final predicted probability p_{t+1} is computed as follows:

$$p_{t+1} = (1 - p_s) * p_k + (1 - p_k) * p_g. \quad (16)$$

Finally, all learnable parameters in ASIKT are trained end-to-end using the Adam optimizer [23] on mini-batches, with the objective of minimizing the binary cross-entropy loss between the predicted probability p_t and the actual answer a_t , as follows:

$$\text{Loss} = \sum_i \sum_t -(a_t^i \log p_t^i + (1 - a_t^i) \log(1 - p_t^i)). \quad (17)$$

5 Experiments

In this section, we conduct a series of experiments to address the following questions:

- **RQ1.** How does ASIKT perform?

Table 1: Statistics of the preprocessed datasets.

	ASSIST09	ASSIST12	ASSIST17	Eedi
# Student	4,116	28,812	1,709	4,918
# Concept	107	254	97	52
# Question	9,798	37,438	2,521	915
# Interaction	302,498	2,652,052	940,175	1,382,542

- **RQ2.** What is the impact of different components in ASIKT?
- **RQ3.** How does ASIKT learn the student's knowledge retrieval process and model changes in their knowledge state? Can ASIKT provide interpretable predictions?
- **RQ4.** How efficient is ASIKT in terms of training and complexity?

5.1 Experimental Setting

5.1.1 Dataset. We evaluated the performance of ASIKT on four commonly used public datasets: ASSIST09¹, ASSIST12², ASSIST17³, and Eedi⁴. The statistical details of the preprocessed datasets are shown in Table 1. During preprocessing, we removed records with missing KC information. To mitigate the impact of data sparsity on question difficulty, we excluded questions that appeared fewer than 10 times and students with fewer than 10 response records. For the Eedi dataset, where a question is associated with multiple hierarchical KCs, we selected only the final-level KC corresponding to each question. The descriptions of the datasets are as follows:

- **ASSIST09:** Collected from the ASSISTments online education platform [16] for the 2009-2010 school year.
- **ASSIST12:** Collected from the ASSISTments online education platform [16] for the 2012-2013 school year.
- **ASSIST17:** Used in the 2017 ASSISTments Data Mining competition, collected from the ASSISTments online education platform [16] for the 2004-2005 and 2005-2006 school years.
- **Eedi:** Used in the 2020 NeurIPS Education Data Mining Challenge, collected from the Eedi online education platform [42] for the period 2018-2020. Due to the large scale of this dataset, we only utilized data from Task 3 and Task 4.

5.1.2 Training Details. In the experiments, all learning sequences were set to a fixed length of 200. Sequences exceeding this length were split into multiple subsequences. For all datasets and models, we applied 5-fold cross-validation. In each fold, 80% of the learning sequences were split into training and validation sets in an 8:2 ratio, while the remaining 20% were used as the test set. During training, the batch size, learning rate, and dropout rate were set to 48, 0.0005, and 0.2, respectively. We employ an early stopping in training with 5 epochs of patience. To prevent overfitting and overparameterization, following AKT [17], we applied the Rasch model to embed questions for the datasets ASSIST09 and ASSIST12, which have a large variety of question types. For the other datasets, we used standard embedding methods. For the models MIKT and PKT, we followed the optimal training settings provided in their

¹<https://sites.google.com/site/assistmentsdata/home/2009-2010-assistment-data>

²<https://sites.google.com/site/assistmentsdata/home/2012-13-school-data-withaffect>

³<https://sites.google.com/view/assistmentsdatamining/dataset>

⁴<https://eedi.com/projects/neurips-education-challenge>

Table 2: Comparison with baseline models. The best results are highlighted in bold, the next best are underlined. * indicates p-value < 0.05 in t-test compared to the second best result.

Datasets	Metrics	DKT	SAKT	AKT	SAINT	simpleKT	DTransformer	PKT	MIKT	ASIKT
ASSIST09	AUC	0.7511	0.7352	0.7864	0.7158	0.7813	0.7552	0.7775	<u>0.7918</u>	0.7978*
	ACC	0.7277	0.7158	0.7443	0.7072	0.7408	0.7263	0.7395	<u>0.7495</u>	0.7519*
ASSIST12	AUC	0.6917	0.6935	0.7560	0.6379	0.7552	0.7444	0.7504	<u>0.7577</u>	0.7589*
	ACC	0.7174	0.7178	0.7432	0.7052	<u>0.7433</u>	0.7383	0.7404	0.7430	0.7438
ASSIST17	AUC	0.7155	0.6498	0.7727	0.6840	0.7577	0.7268	0.7649	<u>0.7758</u>	0.7916*
	ACC	0.6852	0.6629	0.7181	0.6600	0.7089	0.6960	0.7132	<u>0.7212</u>	0.7328*
Eedi	AUC	0.7078	0.7120	0.7737	0.6911	0.7748	0.7684	<u>0.7749</u>	0.7740	0.7754*
	ACC	0.6520	0.6546	0.7061	0.6393	<u>0.7065</u>	0.7014	0.7064	0.7066	0.7066

respective papers. For other baseline methods, we adhered to the best parameter configurations specified in the pyKT library [28]. All models were tuned to their best performance and trained on a Linux server with an Nvidia GeForce RTX 3090 GPU.

5.1.3 Baselines. We compare ASIKT with the following baselines:

- **DKT** [33]: The first model to apply deep learning to knowledge tracing, using RNNs to model student knowledge states.
- **SAKT** [31]: The first model to incorporate attention mechanisms into KT, using attention to capture historical information in the question-solving sequence.
- **AKT** [17]: Embed question difficulty into the question representations based on IRT and uses monotonic attention to track students' knowledge states.
- **SAINT** [10]: Utilizes a transformer architecture to learn both the question sequence and the answer sequence separately.
- **simpleKT** [27]: Simplifies the model structure of AKT while maintaining model performance.
- **DTransformer** [43]: Combines contrastive learning with transformers to track student knowledge states.
- **PKT** [37]: Employs a stage-based approach to model the learning process and explores the impact of question difficulty using IRT.
- **MIKT** [38]: Tracks student knowledge states at both coarse-grained and fine-grained levels, extending the Rasch model to consider the influence of question difficulty.

5.2 Prediction Performance (RQ1)

The prediction performance of the models is shown in Table 2, where the results represent the average performance on the test set across 5-fold cross-validation. We use AUC (Area Under the Curve) and ACC (Accuracy) to evaluate model performance. Based on the results, we observe the following: (1) ASIKT achieves the best performance across all datasets, demonstrating the effectiveness of our proposed model. (2) Compared to recent educational feature-based methods like PKT and MIKT, ASIKT demonstrates superior performance, suggesting that it offers more accurate analysis and modeling of learning sequences, while preserving interpretability. (3) Several classic sequence-based KT methods, such as

Table 3: Ablation study results on four datasets

Models	ASSIST09		ASSIST12		ASSIST17		Eedi	
	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
w/o KSM	0.7888	0.7454	0.7573	0.7424	0.7719	0.7170	0.7705	0.7026
w/o DAKR	0.7575	0.7307	0.7455	0.7385	0.7836	0.7259	0.7709	0.7026
w/o EW	0.7679	0.7320	0.7527	0.7411	0.7858	0.7282	0.7748	0.7055
w/o MD	0.7965	0.7506	0.7587	0.7437	0.7900	0.7305	0.7753	0.7062
w/o SG	0.7962	0.7506	0.7586	0.7434	0.7860	0.7277	0.7751	0.7056
ASIKT	0.7978	0.7519	0.7589	0.7438	0.7916	0.7328	0.7754	0.7066

AKT and simpleKT, show comparable performance to recent educational feature-based methods such as PKT and MIKT, highlighting the strong sequence modeling capabilities of sequence-based approaches. (4) Overall, ASIKT stands out with its strong sequence modeling ability while also maintaining impressive interpretability, outperforming both sequence-based models and educational feature-base models.

5.3 Ablation Study and Sensitivity Analysis (RQ2)

5.3.1 Ablation Study. Through ablation experiments, we analyzed the impact of each key module and setting in ASIKT. As shown in Table 3, ASIKT has five variants: "w/o KSM" replaces the KSM module with a standard Transformer encoder; "w/o DAKR" removes the entire DAKR module; "w/o EW" removes Exercise Weight from DAKR module; "w/o MD" removes Memory Decay from DAKR module; "w/o SG" removes Slipping and Guessing from the Prediction Network. The following observations were made: (1) Replacing the KSM with a Transformer encoder leads to an average performance drop of 1%, highlighting the superior ability of the selective SSM to capture dynamic changes in students' knowledge states. (2) Removing the DAKR module caused a 1.7% average performance decrease. This confirms that powerful sequence modeling methods is crucial for KT models. It also demonstrates that our designed enhanced attention mechanism can accurately model students' knowledge retrieval processes. (3) The exclusion of Exercise Weight leads to a 1.1% performance drop, underscoring its critical role in capturing the student's knowledge retrieval process based on question difficulty. (4) The performance drop from removing Memory Decay and

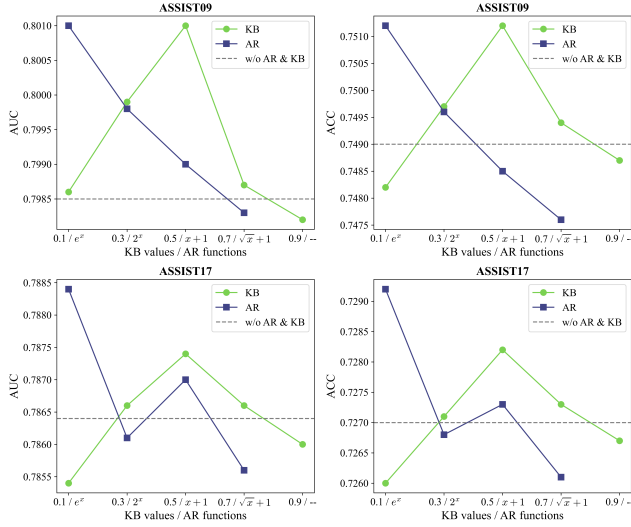


Figure 3: The performance of ASIKT under different values of KB and different functions of AR in Exercise Weight.

Slipping & Guessing was minimal. This aligns with our intuition that students’ forgetting mechanisms are not the primary factor influencing their knowledge states, and Slipping and Guessing behaviors are relatively rare in the final responses.

5.3.2 Sensitivity Analysis of Exercise Weight. To further explore the influence of students’ analogical learning and knowledge bias within Exercise Weight, we conducted a sensitivity analysis on different KB values and AR functions based on the validation set, as shown in Figure 3. Specifically, we evaluated the performance under five different KB values in $\{0.1, 0.3, 0.5, 0.7, 0.9\}$ and four different $AR(x)$ functions in $\{e^x, 2^x, x+1, \sqrt{x}+1\}$. Here, $0 < x < 1$ represents the personalized KC difficulty. When investigating the influence of different KB values, we fixed AR to e^x ; and when studying the impact of various AR functions, we fixed KB to 0.5. “w/o AR & KB” refers to the case where both $AR(x)$ and KB are set to 1, meaning the influence of students’ analogical reasoning and knowledge bias is ignored. Our observations are as follows: (1) When $KB = 0.5$ and $AR(x) = e^x$, the model achieved the best performance on both datasets, which is higher than the “w/o AR & KB” setting, confirming the effectiveness of the experimental setup. (2) Performance declined when KB values were too small or too large, suggesting that a moderate level of attention decay best fits the student’s knowledge bias behavior. (3) The more the $AR(x)$ amplifies the impact of question difficulty, the better the performance on both datasets, indicating that KCs with higher difficulty leave a stronger impression on students during analogical reasoning.

5.4 Interpretability (RQ3)

In this section, we randomly selected students from the test set of the ASSIST09 dataset and conducted a series of experiments based on their learning sequence to illustrate the interpretability of ASIKT.

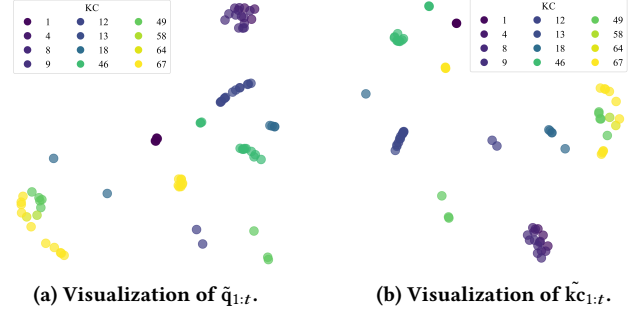


Figure 4: Visualization of the sequence learned by DAKR.

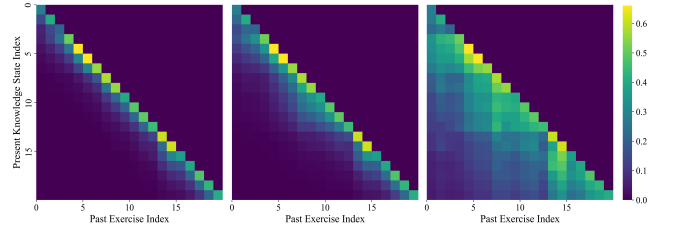


Figure 5: Visualization of weight matrix in the selective SSM of KSM module.

5.4.1 Visualization of the Knowledge Retrieval Process Learned by DAKR. For a student’s learning sequence, we applied t-SNE [39] dimensionality reduction to the learned sequences $\tilde{q}_{1:t}$ and $\tilde{kc}_{1:t}$ from the DAKR module, and randomly selected 12 KCs and their corresponding $\tilde{q}_t$ and $\tilde{kc}_t$ for visualization. As shown in Figure 4, each point represents a $\tilde{q}_t$ and $\tilde{kc}_t$ at a different time step, and points of the same color correspond to the same KC. We observe that, after training with the DAKR module, the points in both sequences are clearly grouped into clusters based on their associated KC. This indicates that the attention mechanism in DAKR, enhanced by Exercise Weight, effectively learns the analogical reasoning patterns of students within the same KC, demonstrating strong interpretability.

5.4.2 Visualization of the Knowledge State Improvement modeled by KSM. Since the SSM within the KSM module employs a matrix-blocking algorithm to improve computational efficiency, the core weight matrix L in equation 14 cannot be directly extracted. To investigate how the SSM captures changes in students’ knowledge states, we designed a dedicated algorithm to reconstruct the weight matrix. As illustrated in Figure 5, we visualized the weight matrices of three different heads in the selective SSM over 20 consecutive time steps from a student’s learning sequence. The values in the figure can be interpreted as the influence weights of historical information on the present knowledge state. It can be observed that the SSM learns the impact of historical exercise information on students’ current knowledge states across different time windows, varying by head. Some heads focus on the influence of short-term historical information, while others attend to long-term historical effects. This demonstrates the effectiveness and interpretability of the selective SSM in modeling students’ knowledge states.

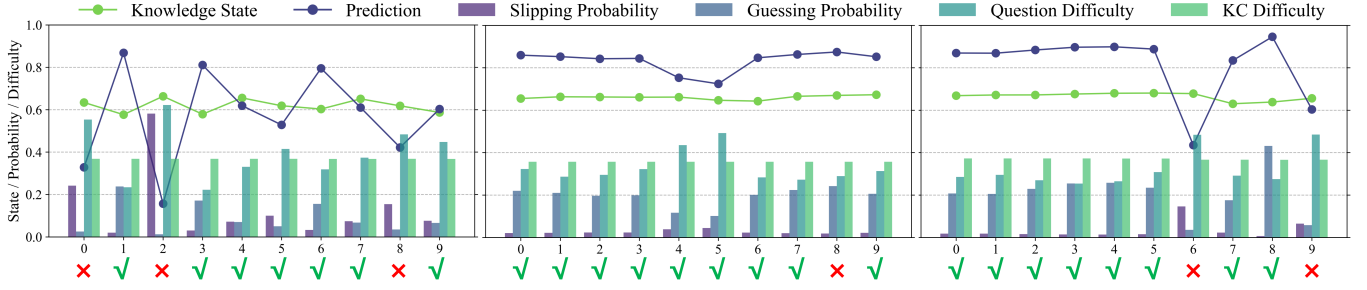


Figure 6: Visualization of ASIKT's prediction process.

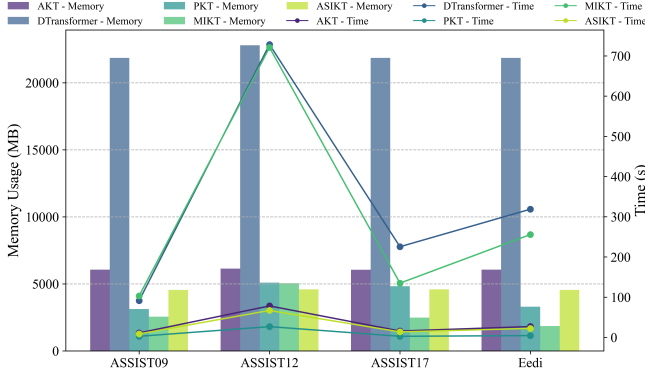


Figure 7: Comparison of training time and memory consumption across different models on four datasets.

5.4.3 Visualization of Students' Performance Prediction Process. ASIKT provides interpretable explanations for the learned knowledge states and final predictions from multiple perspectives, such as question difficulty, KC difficulty, and student behaviors like Slipping and Guessing. We visualized the learning sequences of 10 consecutive time steps from three different students, as shown in Figure 6. The true responses are shown at the bottom. We observe the following: (1) After a correct or incorrect answer, the student's knowledge state increases or decreases accordingly, with stable changes in knowledge state, confirming the model's reasonable representation of knowledge state dynamics. (2) The personalized difficulty of questions and KCs significantly influences the model's predictions. For instance, at time steps with higher question difficulty (e.g., the 0th and 8th steps of the first sequence), the predicted probability decreases, demonstrating ASIKT's ability to capture the impact of difficulty. (3) Besides question and KC difficulty, student behaviors like Slipping and Guessing also affect the model's predictions. For example, at the 2nd step in the first sequence, despite the student's knowledge state being sufficient to solve the problem, the model's predicted probability is lower due to Slipping behavior; at the 7th step in the third sequence, the student's knowledge state has declined, but the model's predicted probability is higher due to Guessing behavior. In conclusion, ASIKT effectively models students' knowledge states and integrates multiple perspectives to provide accurate predictions, demonstrating excellent interpretability.

5.5 Training Time and Memory Consumption (RQ4)

We evaluated the training time and memory consumption of different models on four datasets using a Nvidia GeForce RTX 3090 GPU with 24GB of memory. Due to memory constraints with large batch sizes, we fixed the batch size to 24 and the sequence length to 200 across all models. The average training time and memory consumption for each model over one epoch are shown in Figure 7. We observe that: (1) Compared to Transformer-based sequence models like AKT and DTransformer, especially DTransformer, ASIKT achieves the shortest training time, lowest memory usage, and highest prediction performance. (2) Compared to educational feature-based models like PKT and MIKT, MIKT has the lowest resource usage in most datasets but requires significantly more training time than ASIKT. While PKT has the shortest overall training time, its memory consumption is close to ASIKT's, and its prediction performance is lower. This demonstrates that ASIKT combines the advantages of both approaches, balancing powerful sequence modeling with lower training time and memory usage, achieving an optimal trade-off between performance and simplicity.

6 Conclusion and Future Work

In this paper, we propose ASIKT, an interpretable knowledge tracing model that combines a difficulty-aware attention mechanism with a selective state space model (SSM) to effectively capture students' learning dynamics while ensuring simplicity and interpretability. We first estimate personalized question and KC difficulty from individual learning sequences. Then, we introduce an Exercise Weight and Memory Decay enhanced attention mechanism to model the effects of difficulty and forgetting in the knowledge retrieval process. The selective SSM further captures dynamic knowledge state transitions with reduced model complexity. We also incorporate students' Slipping and Guessing behaviors to enhance interpretability. Experimental results show that ASIKT outperforms state-of-the-art models in both accuracy and interpretability. In future work, we aim to: (1) integrate item response theory to improve embedding robustness, and (2) further refine the selective SSM for better knowledge tracking.

7 Acknowledgments

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